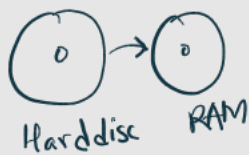
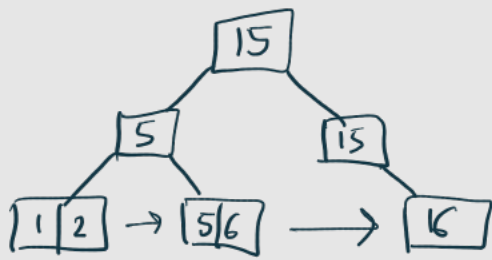


Hashing

① B⁺ tree



* Search : $O(\log n)$

* Sorted (Range)

② Hashing

* Search : $O(1)$

* Not sorted

bucket even

1456	
1298	

--	--

bucket odd

1123	
------	--

id	name	grade	batch
1123			
1456			
1298			

Sum even $\rightarrow$ Even

Sum odd $\rightarrow$ Odd

Question 5:

Suppose you have a table titled "Instructor".

10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

- ✓ A. Make a hash index on attribute instructor_id. Suppose the number of buckets in the hash index is 5 and each bucket can house 2 index entries at max. Bucket overflow is handled using forward chaining.

The hash function, $h = (\text{sum of all digits in id}) \% 5$

B. Explain how to search for the following search keys in your hash index.

- 45565
- 11113

bucket 0

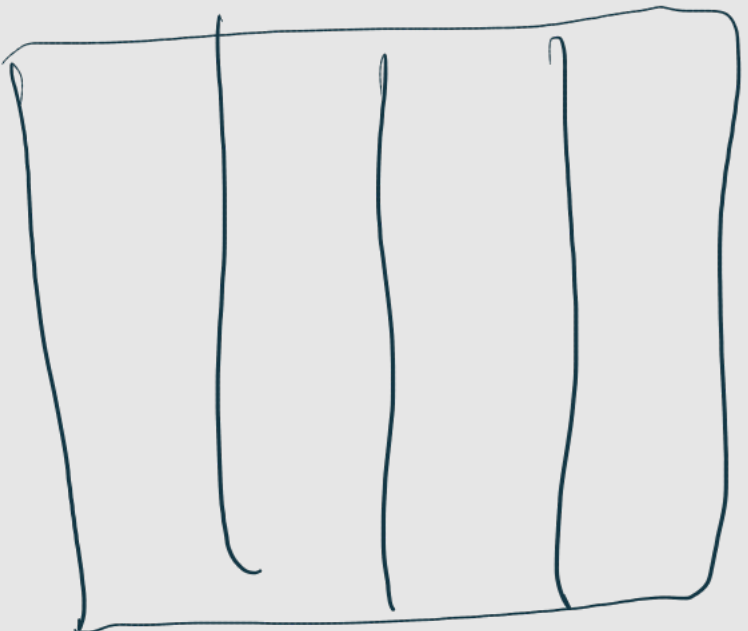
bucket 1

--	--

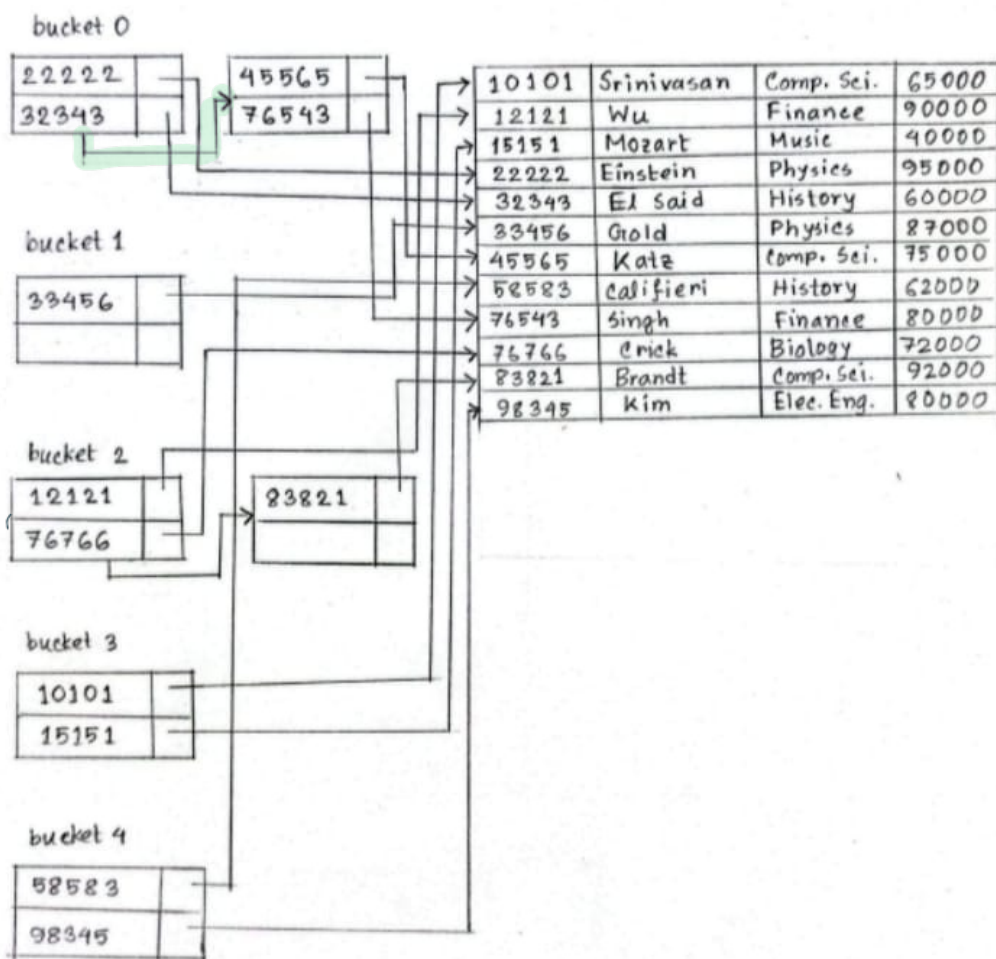
bucket 2

bucket 3

bucket 4



$$h(ID) = (\text{sum of all digits in ID}) \div 5$$



B:

Searching for 45565:

$$h(45565) = 0$$

Therefore, we examine the buckets located at index 0. Upon retrieving this initial bucket, we encounter two entries, namely 22222 and 32343. As our intended record has not been found, we proceed to retrieve the second bucket, from which we obtain the entry 45565.

Searching for 11113:

$$h(11113) = 2$$

Therefore, we examine the buckets located at index 2. Upon retrieving the initial bucket, two records, namely 12121 and 76766, are obtained. Subsequently, retrieving the second bucket yields the entry 83821. At this point, all buckets at index 2 have been retrieved, and the end of the linked list has been reached. Consequently, the search for the key 11113 concludes without success.

More Question:

$$h1 = (\text{First digit of ID}) \div 5$$

$$h2 = (\text{Last digit of ID}) \div 5$$

which is more efficient?

Ans: * overflow कम
* Empty slot कम

Indexing

- (1) $O(\log n)$ search
- ✓ (2) Range search

Hashing

- (1) $O(1)$ search (Fast)
- (2) Not sorted

